

# Supplementary Material for HKVM-RAG: Key-Value-Separated Hypergraph Evidence Organization for Multi-Hop RAG

Mingyu Zhang  
Faculty of Computing  
Harbin Institute of Technology  
Harbin, China  
25b903142@stu.hit.edu.cn

Ying Ma\*  
Faculty of Computing  
Harbin Institute of Technology  
Harbin, China  
y.ma@hit.edu.cn

## I. SCOPE OF THE SUPPLEMENTARY MATERIAL

This supplementary material contains reproducibility and diagnostic evidence that supports, but is not necessary to state, the main paper’s claims. It follows the paper’s bounded framing: HKVM-RAG is evaluated as a fixed-substrate evidence-indexing and dense-aware evidence-control method, not as a hidden-test benchmark, a standalone dense-retrieval replacement, or a full reproduction of every related RAG system. All tables below are derived from frozen prediction files and paper-facing exports. No new LLM extraction calls are introduced by these supplementary analyses.

## II. SOURCE-LEVEL SIGNIFICANCE

Table II gives the paired-bootstrap support for the first-stage source diagnostics summarized in the main paper. The confidence interval tests WHG-KV against the corresponding first-stage baseline. The final column reports the point-estimate margin between WHG-KV and the strongest non-WHG structured complement in the same dataset–first-stage block.

The source-level exports use seeds 13, 29, and 43, 5,000 paper-export bootstrap samples per dataset–first-stage row, frozen train/dev prediction files, and no new LLM extraction calls. The artifact README records the ColBERTv2, BM25, Contriever, and HotpotQA source-level export roots, hashes, and redistributability notes.

## III. DENSE-AWARE CONTROLLER SIGNIFICANCE

Table III expands the main paper’s dense-aware controller significance table. The controller improves over both ColBERTv2 and learned HKVM calibration on all three benchmarks. The 2Wiki gain over learned HKVM is intentionally small but positive, because learned HKVM calibration and the HKVM-only controller are already strong on that dataset.

For HotpotQA, EM paired-bootstrap intervals mirror the F1 result: the controller improves over ColBERTv2 by +5.991 EM with 95% CI [+5.510, +6.482], and over learned HKVM by +2.921 EM with 95% CI [+2.579, +3.255].

Controller bootstrap exports use 5,000 samples over frozen prediction files. MuSiQue and 2Wiki use the OOF-confirmed

Phase1D export; HotpotQA uses the matched 20260602 dense-aware controller export. The source-level HotpotQA ColBERTv2 row in Table II is a separate source-replacement export, which explains its slightly different confidence interval from the main controller-vs-ColBERTv2 interval in Table III.

## IV. EVIDENCE-SELECTION TRAJECTORY

Table IV records the trajectory from no-gold WHG retrieval to oracle answer-path selection, train-derived support scoring, learned HKVM calibration, and the dense-aware controller. This table is a diagnostic trajectory rather than a single deployed pipeline. The oracle rows use support information and serve only to measure headroom.

The trajectory combines fixed-substrate dev outputs, oracle diagnostics, train-to-dev support-scoring outputs, Phase1C learned HKVM calibration, and Phase1D dense-aware controller exports. It should be read as a mechanism-oriented evidence path rather than as a single chronological production pipeline.

## V. STANDALONE BOUNDARY ON HOTPOTQA

HotpotQA is included to prevent a standalone-retrieval overclaim. In the fixed-substrate standalone matrix, WHG-KV improves support coverage over KG-PPR but does not improve answer F1. Table V separates that standalone boundary from the controller-mediated recovery reported in the main paper.

This boundary is compatible with the dense-aware controller result: raw structured retrieval can underperform lexical and dense baselines on HotpotQA, while WHG-KV rank/score signals can still help a learned controller reorder answer-bearing passages.

## VI. NEGATIVE AND BOUNDARY DIAGNOSTICS

### A. Naive Fusion

MuSiQue contains genuine ColBERTv2–WHG complementarity, but simple fixed-budget fusion and shallow single-feature gates did not reliably exceed ColBERTv2. The best conservative fixed-budget mixture improved all-gold coverage

TABLE I

SUPPLEMENTARY EVIDENCE MAP. THE SUPPLEMENT RECORDS STATISTICAL SUPPORT, BOUNDARY DIAGNOSTICS, AND ARTIFACT PROVENANCE THAT WOULD OVERBURDEN THE MAIN PAPER; IT DOES NOT INTRODUCE ADDITIONAL BENCHMARK CLAIMS BEYOND THE FROZEN RESULT MATRIX.

| Supplement section            | Supports                                                                 | Does not support                                                    |
|-------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------|
| Source-level significance     | WHG-KV-specific complementarity under the tested first-stage sources     | Generic benefit from arbitrary structured sources                   |
| Controller significance       | Dense-aware controller gains over ColBERTv2 and learned HKVM calibration | Hidden-test or deployment-throughput claims                         |
| Evidence-selection trajectory | Support-selection bottleneck and learned recovery path                   | Treating oracle rows as deployable systems                          |
| HotpotQA boundary             | Separation of standalone failure and controller recovery                 | Standalone WHG-KV superiority on HotpotQA                           |
| Negative diagnostics          | Need for learned control and cache-manifest discipline                   | Claim that LLM triples dominate all OpenIE alternatives             |
| Artifact provenance           | Audit path for paper-facing result families                              | Redistribution of non-redistributable provider or dataset artifacts |

TABLE II

SOURCE-LEVEL BASELINE-VS-WHG PAIRED-BOOTSTRAP SUPPORT AND NON-WHG MARGINS. THE CONFIDENCE INTERVAL TESTS WHG-KV AGAINST THE CORRESPONDING FIRST-STAGE BASELINE; THE FINAL COLUMN IS A POINT-ESTIMATE MARGIN OVER THE STRONGEST MATCHED NON-WHG SOURCE. THE TABLE DOES NOT IMPLY THAT ALL STRUCTURED SOURCES HELP OR THAT THE DIAGNOSTICS COVER ALL RETRIEVER FAMILIES.

| Dataset         | First-stage | WHG-KV $\Delta$ F1 vs base | 95% CI             | Margin over best non-WHG |
|-----------------|-------------|----------------------------|--------------------|--------------------------|
| MuSiQue         | ColBERTv2   | +6.763                     | [+5.813, +7.753]   | +8.219                   |
| MuSiQue         | BM25        | +12.038                    | [+10.935, +13.120] | +12.212                  |
| MuSiQue         | Contriever  | +7.813                     | [+6.810, +8.814]   | +7.605                   |
| 2WikiMultiHopQA | ColBERTv2   | +11.084                    | [+10.722, +11.442] | +8.077                   |
| 2WikiMultiHopQA | BM25        | +16.452                    | [+16.039, +16.865] | +9.786                   |
| 2WikiMultiHopQA | Contriever  | +23.119                    | [+22.668, +23.577] | +8.845                   |
| HotpotQA        | ColBERTv2   | +5.966                     | [+5.479, +6.456]   | +5.397                   |
| HotpotQA        | BM25        | +4.971                     | [+4.554, +5.374]   | +4.188                   |
| HotpotQA        | Contriever  | +5.166                     | [+4.683, +5.634]   | +5.522                   |

TABLE III

DENSE-AWARE CONTROLLER SIGNIFICANCE. THE TABLE SUPPORTS THE MAIN CONTROLLER CLAIM AGAINST COLBERTV2 AND LEARNED HKVM CALIBRATION; IT DOES NOT CONVERT THE CACHED TRAIN-TO-DEV PROTOCOL INTO A HIDDEN-TEST BENCHMARK.

| Dataset  | Comparator   | $\Delta$ F1 | 95% CI             |
|----------|--------------|-------------|--------------------|
| MuSiQue  | ColBERTv2    | +6.763      | [+5.791, +7.726]   |
| MuSiQue  | Learned HKVM | +8.319      | [+7.327, +9.293]   |
| 2Wiki    | ColBERTv2    | +11.084     | [+10.740, +11.437] |
| 2Wiki    | Learned HKVM | +0.455      | [+0.368, +0.542]   |
| HotpotQA | ColBERTv2    | +5.966      | [+5.491, +6.452]   |
| HotpotQA | Learned HKVM | +2.881      | [+2.555, +3.201]   |

but reduced answer F1. This motivates the learned passage-level controller used in the main paper, and blocks the simpler claim that generic RRF or quota fusion is enough.

### B. Extraction Substrate

The shared LLM-triple cache is a controlled extraction substrate, not a claim that LLM triples dominate all OpenIE alternatives. In extraction-quality diagnostics, LLM triples were much sparser than heuristic OpenIE: objects per passage changed from 34.814 to 4.249 on 2Wiki and from 47.056 to 4.878 on MuSiQue. Pairwise KG-PPR with the sparse LLM tuple cache did not dominate heuristic OpenIE KG-PPR in that diagnostic. The paper therefore treats cache changes as new manifests rather than mixing them into the frozen result matrix.

## VII. RUN AND ARTIFACT PROVENANCE

The public artifact is split across two hosting platforms:

- **GitHub** (code, configs, scripts, results, manifests): <https://github.com/Mingyu-Zh/HKVM-RAG>

- **Hugging Face Dataset** (data, frozen outputs): <https://huggingface.co/datasets/MingY-Zh/HKVM-RAG>

This supplement itself is hosted in the GitHub repository at `supplemental_material.pdf`. Reviewers should download `data/` and `frozen_outputs/` from Hugging Face and place them at the repository root before running the paper-evidence verification script.

Table VI lists the paper-facing result families and the artifact records required to verify them. The artifact includes scripts (`scripts/verify_paper_evidence.py`, `scripts/reproduce_fixed_substrate.py`), configs (`configs/`), and a SHA-256 file manifest (`manifests/FILES.sha256`) for integrity verification.

TABLE IV

EVIDENCE-SELECTION TRAJECTORY FROM NO-GOLD WHG RETRIEVAL TO DENSE-AWARE CONTROL. THE TABLE SUPPORTS THE INTERPRETATION THAT SUPPORT SELECTION IS A REPAIRABLE BOTTLENECK; ORACLE ROWS ARE DIAGNOSTIC UPPER BOUNDS AND NOT DEPLOYABLE MODEL RESULTS.

| Variant                      | MuSiQue |        |        | 2WikiMultiHopQA |        |         | HotpotQA |        |         |
|------------------------------|---------|--------|--------|-----------------|--------|---------|----------|--------|---------|
|                              | F1      | EM     | AR@10  | F1              | EM     | AR@10   | F1       | EM     | AR@10   |
| No-gold WHG retrieval        | 39.925  | 39.222 | 38.643 | 79.299          | 78.650 | 99.650  | 72.957   | 72.586 | 98.190  |
| Oracle answer-path selection | 74.330  | 74.059 | 73.066 | 90.884          | 90.832 | 99.634  | 91.111   | 91.047 | 98.190  |
| Train-derived support scorer | 54.252  | 53.662 | 50.228 | 87.212          | 86.967 | 98.378  | 81.525   | 81.269 | 96.016  |
| Learned HKVM calibration     | 56.754  | 56.213 | 53.303 | 88.391          | 88.181 | 99.030  | 82.929   | 82.705 | 96.439  |
| Dense-aware HKVM controller  | 65.073  | 64.722 | 64.115 | 88.846          | 88.658 | 100.000 | 85.810   | 85.627 | 100.000 |

TABLE V

HOTPOTQA STANDALONE BOUNDARY UNDER MATCHED CANDIDATE-PASSAGE BUDGETS. THE TABLE BLOCKS A STANDALONE WHG-KV SUPERIORITY CLAIM WHILE REMAINING COMPATIBLE WITH CONTROLLER-MEDIATED RECOVERY.

| Method         | AR@10   | F1     | EM     |
|----------------|---------|--------|--------|
| BM25           | 99.851  | 81.835 | 81.648 |
| Contriever     | 100.000 | 79.930 | 79.730 |
| ColBERTv2      | 100.000 | 79.844 | 79.635 |
| KG-PPR         | 95.381  | 73.645 | 73.315 |
| Static-HG      | 98.190  | 73.340 | 73.018 |
| Weighted-HG-KV | 98.190  | 72.957 | 72.586 |

**TABLE VI**  
PAPER-FACING RESULT PROVENANCE. THE TABLE IDENTIFIES THE PROTOCOL AND ARTIFACT RECORD FOR EACH RESULT BLOCK; EXACT EXPORT ROOTS, HASHES, AND REDISTRIBUTABILITY CONSTRAINTS BELONG IN THE SUBMITTED ARTIFACT README.

| Result block                  | Verifies                                                              | Prediction protocol                                                             | Bootstrap/export                                                            | Artifact record                                                          |
|-------------------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------|
| P3 fixed-substrate matrix     | Standalone WHG-vs-KG comparison and HotpotQA boundary                 | Frozen dev predictions; shared tuple cache; seeds 13/29/43                      | Seed-13 1,000-sample WHG-minus-KG bootstrap; paper table export             | results/paper_evidence/main_fixed_frozen_outputs/predictions/fixed_s     |
| Evidence-selection trajectory | Oracle headroom, support scorer, calibration, and controller sequence | Frozen dev predictions plus train-to-dev models; no dev labels at inference     | Phase-specific paper exports; controller rows use 5,000-sample bootstrap    | results/paper_evidence/calibration<br>results/run_summaries/             |
| Phase1D controller            | Dense-aware controller vs ColBERTv2 and learned HKVM                  | OOF train-side HKVM predictions; frozen dev predictions; seeds 13/29/43         | 5,000-sample paired bootstrap over frozen prediction files                  | results/paper_evidence/adaptive_so<br>frozen_outputs/predictions/adaptiv |
| Source-level diagnostics      | WHG-KV source specificity under ColBERTv2/BM25/Contriever             | First-stage source fixed per block; structured source swapped; no new LLM calls | 5,000-sample baseline-vs-WHG bootstrap; non-WHG margins are point estimates | results/paper_evidence/source_leve<br>bootstrap CSVs                     |
| Negative diagnostics          | Naive-fusion and extraction-substrate boundaries                      | Diagnostic exports outside the frozen main matrix                               | Descriptive unless explicitly reported as paired bootstrap                  | results/run_summaries/extraction_q<br>prompt/schema records              |